

NEUHAUSEN OB ECK, Germany

WMO: 109210

Lat: 47.983N

Lon: 8.900E

Elev: 2638

StdP: 13.35

Time Zone: 1.00 (EUC)

Period: 84-93

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 5.2	(c) 10.5	(d) 3.4	(e) 7.2	(f) 6.4	(g) 8.4	(h) 9.3	(i) 12.4	(j) 24.1	(k) 36.5	(l) 21.8	(m) 35.3	(n) 7.1	(o) 70	(p) 0.585

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.5	80.5	64.2	77.1	62.8	74.8	61.5	65.9	77.0	63.9	74.4	62.3	72.5	7.1	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.2	89.0	71.3	59.4	83.4	68.9	57.6	78.2	67.0	32.1	77.5	30.6	74.2	29.3	72.4	73.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 18.8	(b) 16.5	(c) 14.4	(e) 1.8	(f) 86.2	(g) 9.3	(h) 2.0	(i) -4.8	(j) 87.7	(k) -10.2	(l) 88.9	(m) -15.4	(n) 90.0	(o) -22.2	(p) 91.5		
			DB	WB	DB	WB	DB	WB	DB	WB	DB	WB	DB	WB		
			1.8	86.2	9.3	2.0	-4.8	87.7	-10.2	88.9	-15.4	90.0	-22.2	91.5		
			1.8	69.4	9.3	2.0	-4.9	70.9	-10.4	72.1	-15.6	73.2	-22.4	74.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	44.9	28.2	29.7	37.1	43.1	51.1	55.8	61.5	61.7	55.5	47.2	35.8	31.5
	DBStd	13.84	9.49	8.70	7.88	6.81	7.38	6.32	5.76	6.13	6.00	7.10	7.07	7.48
	HDD50	3130	675	569	405	223	81	17	1	2	17	139	426	575
	HDD65	7395	1140	989	865	656	432	283	137	137	289	551	876	1040
	CDD50	1284	0	0	5	17	114	190	358	364	182	53	1	0
	CDD65	73	0	0	0	0	1	6	30	33	3	0	0	0
	CDH74	707	0	0	0	0	8	57	278	333	29	2	0	0
	CDH80	108	0	0	0	0	0	2	47	59	0	0	0	0
	WSAvg	6.4	6.5	6.7	7.2	6.9	6.8	6.5	6.1	5.8	6.0	5.9	6.0	6.0
	PrecAvg	33.4	2.2	2.0	2.0	2.3	3.6	3.8	4.0	3.3	2.6	2.6	2.4	2.6
Precipitation	PrecMax	39.7	5.7	5.1	6.8	4.8	6.2	6.5	7.3	6.4	4.4	4.9	5.2	6.3
	PrecMin	25.4	0.2	0.5	0.5	0.1	1.0	1.6	1.8	0.9	0.9	0.2	0.0	0.0
	PrecStd	4.1	1.5	1.2	1.3	1.2	1.4	1.2	1.3	1.3	1.0	1.3	1.3	1.5
	PrecStd	4.1	1.5	1.2	1.3	1.2	1.4	1.2	1.3	1.3	1.0	1.3	1.3	1.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	50.1	56.8	66.0	69.5	75.2	79.1	84.6	85.8	78.4	72.0	56.9	51.9
		MCWB	44.9	45.9	51.6	53.5	59.7	63.6	65.5	66.5	63.3	58.5	49.6	44.4
	2%	DB	46.2	48.6	57.5	64.7	70.2	75.2	80.4	80.9	73.7	66.4	51.9	48.0
		MCWB	42.2	42.2	47.0	51.4	56.9	61.9	64.6	63.6	60.9	56.1	46.8	43.2
	5%	DB	42.7	44.6	53.2	60.4	67.8	71.6	77.0	77.4	70.9	61.2	48.4	44.6
		MCWB	40.1	39.9	44.5	48.9	55.8	59.7	63.0	62.7	59.4	53.9	45.2	41.6
	10%	DB	39.4	40.9	48.3	55.5	64.3	67.9	73.5	74.8	66.5	57.3	46.0	41.3
		MCWB	37.5	38.2	42.5	46.5	53.5	57.5	61.8	61.5	57.5	51.8	43.5	39.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	45.6	46.8	51.7	54.3	61.4	66.7	68.7	68.2	64.7	59.3	51.2	47.3
		MCDB	49.5	55.1	64.5	66.9	71.6	76.3	79.8	81.3	75.1	69.7	56.3	50.6
	2%	WB	42.5	42.8	47.7	52.0	58.3	63.0	66.2	65.6	62.1	56.8	48.1	43.7
		MCDB	44.8	47.3	56.9	63.5	68.5	72.7	76.6	77.9	71.9	65.2	50.5	46.8
	5%	WB	39.9	40.4	45.3	49.6	56.4	60.6	64.3	63.6	60.1	54.7	45.6	41.4
		MCDB	41.9	43.8	51.6	58.7	65.8	70.4	74.6	74.9	68.6	61.0	48.0	43.6
	10%	WB	37.4	37.9	43.0	47.3	54.6	58.4	62.5	62.0	58.3	52.3	43.2	39.1
		MCDB	38.8	40.6	47.0	54.4	62.9	66.3	71.9	72.7	65.3	57.2	45.3	40.8
Mean Daily Temperature Range	5% DB	MDBR	11.2	12.9	14.6	16.2	17.2	17.1	18.5	19.3	17.2	15.0	11.7	10.5
		MCDBR	14.8	20.0	22.6	24.7	22.8	22.3	23.9	24.8	22.6	22.0	16.9	15.9
		MCWBR	11.8	14.0	13.9	14.6	13.0	11.9	12.9	12.2	12.6	13.7	13.0	11.9
	5% WB	MCDBR	13.0	17.0	19.8	22.8	20.3	20.5	21.3	22.5	20.6	21.2	15.1	13.6
		MCWBR	10.7	12.6	12.5	14.1	12.4	11.5	12.1	11.8	12.0	13.6	12.3	10.7
Clear-Sky Solar Irradiance	taub		0.281	0.304	0.335	0.366	0.377	0.381	0.382	0.377	0.354	0.330	0.302	0.278
	taud		2.473	2.416	2.385	2.325	2.326	2.342	2.330	2.362	2.411	2.470	2.502	2.474
	Ebn at Noon		257	273	280	281	281	279	277	274	270	259	246	243
	Edh at Noon		20	26	32	37	39	38	38	36	31	25	19	18
All-Sky Solar Radiation	RadAvg		335	589	970	1358	1584	1816	1727	1510	1124	696	378	272
	RadStd		40	88	135	194	164	158	155	152	118	75	47	45

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (108 neighbors)	N/A	N/A	N/A	N/A	+0.76	+0.67	N/A	N/A	+92	+20

Nomenclature: See separate page